THE CHALLENGE · RECOVER THE SECRET SENT OVER RADIO

The scale was connected. And a little too chatty.

RADIO RECORDING

10 seconds of signal

Exchanges between the scale and gateway.

ESP32-C6 FIRMWARE

The scale's program

It reveals the protocol and its rules.

LAYER 1 · RADIO

Signal → packets — Recover the transmitted bytes.

LAYER 2 · ZIGBEE NETWORK

Packets → reports — Find the network key in the pairing exchange.

LAYER 3 · APPLICATION

Fragments → flag — Rebuild the key tied to this scale.

THE TAKEAWAY

Three separate steps: **decode, decrypt the network, open the fragments.**

STEP 1 · DECODE THE SIGNAL

Before decrypting, we need to recover the packets.

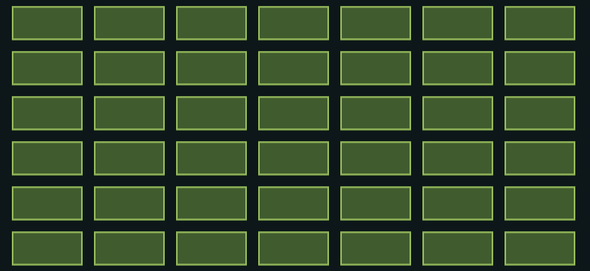
RADIO SIGNAL · ILLUSTRATION



Recognise the patterns used by Zigbee.

→

RECOVERED PACKETS



42 / 42

Valid error checks.

THE TAKEAWAY

The error check confirms **that the bytes were decoded correctly.**

STEP 2 · LISTEN AS THE DEVICE JOINS

When it joins the network, the scale receives a key.

THE PAIRING EXCHANGE IS IN THE CAPTURE

The gateway sends a network key, protected using a known default key.

DEFAULT KEY

`ZigBeeAlliance09`

→

ZIGBEE RULE

Derive a transport key

→

CAPTURED MESSAGE

Open “Transport-Key”

VERIFIED RESULT

Network key recovered ✓

The message authenticity check passes.

→

DECRYPTED REPORTS

6 fragments · 46 bytes

Still encrypted by the application.

THE TAKEAWAY

The network key becomes accessible **through its recorded distribution exchange.**

STEP 3 · TIE THE MESSAGE TO THIS SCALE

The final key depends on the scale's identity.

RADIO IDENTIFIER · OBSERVED EUI-64

`40:4c:ca:ff:fe:46:87:c4`

+

FIRMWARE CONSTANT

“propolis”

+

FRAGMENT NUMBER

0, 1, 2, 3, 4 or 5

```
[+] network key: b8d8e46f82f99446568d2d30e9512c16
[+] balance EUI-64 (LE on air): c48746feffca4c40  = 40:4c:ca:ff:fe:46:87:c4
[+] fragments: {0: '9f06a697fe855673', 1: '450121232be6ef6f', 2: '72e46c164c1d7da4', 3:
'f23cd1e81057e688', 4: 'd51eb01f236057ca', 5: '73c2ed833409'}
[+] FLAG: BA26{l3s_4b31ll3s_n3_s0nt_j4m41s_s1l3nc13us3s}
```

the recipe = the flag

SAVED EVIDENCE · CLICK TO ENLARGE

Rebuild the key for each fragment.

Apply RC4, then assemble in order.

The identity makes the key device-specific; it is visible in the capture.

`BA26{l3s_4b31ll3s_n3_s0nt_j4m41s_s1l3nc13us3s}`

THE TAKEAWAY

The solve **rebuilds the keys** from known data, then decrypts with RC4.